Xiaojuan Gou

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EDUCATION

- **The University of New South Wales**

  Visiting Ph.D. researcher in Biotech & Biomolecular Science — Sydney, Australia

  Supervisor : Associate Professor Emily Wong — *Apr. 2025 – Present*

- **Huazhong Agricultural University**

  Ph.D. Candidate (M.S. – Ph.D. Program) in neurobiology and immunology — Wuhan, China

  Supervisor : Prof. Gang Cao & Prof. Xiangru Wang — *Sep. 2020 – Present*

- **Sichuan Agricultural University**

  B.S. in Veterinary Medicine — Sichuan, China

  GPA: 4.06/5 (Rank: 1/220) — *Sep. 2016 – Jun. 2020*

RESEARCH EXPERIENCE

- **Evolutionary convergence and divergence of hippocampal cytoarchitecture between rodents and primates revealed by single-cell spatial transcriptomics (Primary Collaborative Project)**

  Integrated analysis of single-cell transcriptome and spatial transcriptome data to draw a spatial transcriptome map of the hippocampus of three species.

  (Jun. 2022 – Jan. 2026)

- **Transport stress induces neuro-immune dysregulation and exacerbates Mycobacterium tuberculosis infection in mice (PhD Thesis Project)**

  Use single cell and bulk transcriptome data to study the regulatory mechanisms between nerves and immunity.

  (Apr. 2023 – Feb. 2026)

- **Strategies to package recombinant Adeno-Associated Virus expressing the N-terminal gasdermin domain for tumor treatment**

  Using single-cell transcriptome data analysis to explain rAAV-P2 treatment activates antitumor immunity and inhibits the growth of triple-negative breast cancer. *(Oct. 2020 – Dec. 2021)*

- **Single-cell transcriptome landscape and antigen receptor dynamic during SARS-CoV-2 vaccination** *(Feb. 2022 – Jul. 2023)*

  Mapping single-cell profiles of immune responses following vaccination with an inactivated COVID-19 vaccine.

- **A single-cell transcriptomic atlas of mouse prefrontal cortex maps dynamics of intratelencephalic neurons during postnatal development**

  Assisted in single-cell RNA-seq analysis for neural circuit research, supporting the construction of spatiotemporal atlases of mouse PFC development.

  (Feb. 2024 – Jan. 2026)

- **Identification of Zebrafish Ortholog for Human EC1.45 Enhancer**

  Identified the zebrafish ortholog of the human EC1.45 enhancer and analyzed its conserved regulatory features using sequence alignment, TF motif analysis, machine learning, and genome synteny.

  (Jun. 2025 – Present)

PUBLICATIONS

Gou X, Lei Y, Yang B, Zhou W, Chen X, Yan K, Zhang Q, Zheng H, Deng L, Guo T, et al. Transport Stress Induces Neuroimmune Dysregulation and Exacerbates Mycobacterium Tuberculosis Infection in Mice. ***International Journal of Molecular Sciences***. 2026; 27(4):1919. https://doi.org/10.3390/ijms27041919

Fei, T., Dong, Y., **Gou, X.,** Zhang, J., Wang, G., Xu, C., et al. (2026). Evolutionary convergence and divergence of hippocampal cytoarchitecture between rodents and primates revealed by single-cell spatial transcriptomics. ***National Science Review***, https://doi.org/10.1093/nsr/nwaf595.

Zheng H., Yan, K., **Gou, X.,** Wang, Z., Yang, L., Huang, Y., Liu, H., Dai, J., Sun, L., Cao, G. (2026). A single-cell transcriptomic atlas of mouse prefrontal cortex maps dynamics of intratelencephalic neurons during postnatal development. ***PLOS Biology***, https://doi.org/10.1371/journal.pbio.3003594.

Sun, L., Zheng, H., Huang, Y., Huang, X., Yan, K., Wang, Z., Yang, L., Yue, Y., **Gou, X**., Du, G., Wang, Y., Cao, G. (2026). Organization of mouse prefrontal cortex subnetwork revealed by spatial single-cell multiomic analysis of SPIDER-Seq. ***National Science Review***. https://doi.org/10.1093/nsr/nwag004.

Cao, X., Chen, X., Zhu, Y., **Gou, X.,** Yan, K., Yang, B., Men, D., Liu, L., Zhang, Y.A., and Cao, G. (2022). Single-cell transcriptome landscape and antigen receptor dynamic during SARS-CoV-2 vaccination. ***Genes & diseases*** 10, 1675-1686. https://doi.org/10.1016/j.gendis.2022.08.020.

Lu, Y., He, W., Huang, X., He, Y., **Gou, X.,** Liu, X., Hu, Z., Xu, W., Rahman, K., Li, S., et al. (2021). Strategies to package recombinant Adeno-Associated Virus expressing the N-terminal gasdermin domain for tumor treatment. ***Nature communications*** 12, 7155. https://doi.org/10.1038/s41467-021-27407-0.

RESEARCH INTERESTS

- Omics data analysis
- Single-cell and spatial omics for studying neuro-immune-infection dynamics
- Software development for data analysis

SKILLS

- **Languages:** English (IELTS : 6.5)
- **Omics data analysis:** RNA-seq, single-cell and spatial transcriptomics, Hi-C, ChIP-seq, ATAC-seq, CUT&RUN, and other related techniques
- **Programming & tools:** Python, R, Linux, SQL, common bioinformatics packages
- **Computational biology:** Pipeline development, data visualization, statistical modeling

REFEREE DETAILS

Professor Xiangru Wang - PhD supervisor
Huazhong Agricultural University
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Associate Professor Emily Wong - Host PhD supervisor
The University of New South Wales
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Professor Gang Cao - Master's supervisor
Shenzhen University of Advanced Technology
caog@siat.ac.cn